

Mini Study Guide 2: Short Sales and Margin Calls

FIN 4000 Investments · Midterm prep · BKM Ch. 3

Learning objectives

A short seller gets a margin call when the stock price *rises* far enough that equity falls below the maintenance margin. Most exam errors come from reusing the long-position formula.

After this guide you should be able to:

1. Set up a short-sale margin account: assets, liability, and equity.
2. Compute the margin percentage at any price.
3. Solve for the price that triggers a margin call.
4. Compute the cash needed to meet a margin call.
5. Compute the return on a short sale, including dividends the short seller must pay.

Core concept: the short-sale account

To sell short, you borrow shares from your broker, sell them, and must later buy them back to return them. The sale proceeds stay in your account; you cannot withdraw them. You also post initial margin (cash or T-bills).

Account item	Amount	Changes when price moves?
Asset: sale proceeds	$n \times P_0$	No (fixed)
Asset: margin deposit	$IM \times n \times P_0$	No (unless you add cash or pay dividends)
Liability: shares owed	$n \times P$	Yes (rises when P rises)
Equity	Total assets – $n \times P$	Falls when P rises

Margin percentage on a short position:

$$\text{Margin} = \frac{\text{Equity}}{\text{Value of shares owed}} = \frac{(\text{Proceeds} + \text{Deposit}) - nP}{nP}$$

Where:

- Equity = account assets minus the current value of the shares owed
- Proceeds = $n \times P_0$, the cash received from the short sale
- Deposit = the initial margin posted, $IM \times n \times P_0$
- n = number of shares sold short
- P = current share price

Margin call price. Set margin equal to the maintenance margin (MM) and solve for P:

$$P^* = \frac{\text{Proceeds} + \text{Deposit}}{n(1 + MM)} = \frac{P_0(1 + IM)}{1 + MM}$$

Where:

- P^* = the share price that triggers a margin call
- Proceeds, Deposit, n = as defined above
- MM = maintenance margin, as a decimal
- P_0 = price at which the shares were sold short

- IM = initial margin, as a decimal
- Loan (long position, table below) = amount borrowed from the broker

Long vs. short margin: side by side.

	Buying on margin (long)	Short sale
You lose when	Price falls	Price rises
Liability	Broker loan (fixed dollars)	Shares owed ($n \times P$, moves with price)
Equity	$nP - \text{Loan}$	$(\text{Proceeds} + \text{Deposit}) - nP$
Margin ratio	Equity / nP	Equity / nP
Margin call when	$P \text{ falls to } \text{Loan} / [n(1 - \text{MM})]$	$P \text{ rises to } (\text{Proceeds} + \text{Deposit}) / [n(1 + \text{MM})]$
Maximum loss	Initial equity plus the loan (price goes to 0)	Unlimited in theory
Dividends	You receive them	You pay them to the share lender

Worked examples

Example 1: Margin call price (Algorithmic)

You short 1,000 shares at \$100. Initial margin is 50% and maintenance margin is 30%.

1. Proceeds = $1,000 \times \$100 = \$100,000$.
2. Deposit = $0.50 \times \$100,000 = \$50,000$.
3. Total assets = \$150,000 (fixed).
4. Margin call price: $P^* = \$150,000 / [1,000 \times (1 + 0.30)] = \mathbf{\$115.38}$.

Check: at \$115.38, shares owed = \$115,385 and equity = $\$150,000 - \$115,385 = \$34,615$.
Margin = $34,615 / 115,385 = 30\%$.

BA II Plus: $150000 \div 1.3 \div 1000 = \rightarrow 115.38$

Example 2: Meeting the margin call (Algorithmic)

The stock in Example 1 rises to \$120.

1. Shares owed = \$120,000. Equity = $\$150,000 - \$120,000 = \$30,000$.
2. Margin = $30,000 / 120,000 = \mathbf{25\%}$, which is below 30%: margin call.
3. Equity required at 30% maintenance = $0.30 \times \$120,000 = \$36,000$.
4. Cash to deposit = $\$36,000 - \$30,000 = \mathbf{\$6,000}$.

Brokers sometimes require you to restore margin to the initial level instead. At 50%, you would need $0.50 \times \$120,000 - \$30,000 = \$30,000$. Read the question for which level applies.

Example 3: Dividends and the return on a short sale (Algorithmic)

You short 500 shares at \$40 with 50% initial margin (\$10,000 deposit) and 25% maintenance margin. The stock pays a \$1.00 dividend while you are short.

1. Margin call price with no dividend: $\$30,000 / (500 \times 1.25) = \mathbf{\$48.00}$.
2. The \$500 dividend is paid out of your account, so total assets fall to \$29,500. New margin call price: $\$29,500 / 625 = \mathbf{\$47.20}$. The dividend brings the margin call closer.
3. You cover at \$44. Loss on shares = $(44 - 40) \times 500 = \$2,000$. Plus dividend = \$500. Total loss = \$2,500.
4. Return on initial margin = $-\$2,500 / \$10,000 = \mathbf{-25\%}$.

If instead you cover at \$34: gain = $(40 - 34) \times 500 - \$500 = \$2,500$, a return of **+25%**.

$$r_{\text{short}} = \frac{n(P_0 - P_1) - n \times D}{\text{Initial margin deposit}}$$

Where:

- $r(\text{short})$ = return on the short position, measured on the margin deposit
- n = number of shares sold short
- P_0 = price at which the shares were sold
- P_1 = price at which the position is covered
- D = dividend per share paid while the position is open
- Initial margin deposit = cash posted when the position was opened

Example 4: Compare with a long margin purchase

You buy 1,000 shares at \$100 with 50% initial margin (a \$50,000 loan) and 30% maintenance margin.

- Margin call price = $\text{Loan} / [n(1 - \text{MM})] = \$50,000 / (1,000 \times 0.70) = \mathbf{\$71.43}$.
- The long position is called on a 28.6% price *drop*. The short in Example 1 is called on a 15.4% price *rise*.

Common pitfalls

- **Using the long formula.** A short's margin call comes on a price *rise*. If your answer is below P_0 , recheck.
- **Forgetting the proceeds.** Total assets = proceeds + deposit, not the deposit alone.
- **Dividing by the wrong base.** Margin = equity / *current* value of shares owed ($n \times P$), not initial value.
- **Dropping dividends.** The short seller pays them. They lower the return and the margin call price.
- **Return base.** Return is measured on the initial margin deposit, not on the proceeds.

Key terms and definitions

- ☐ **Short sale:** selling borrowed shares in the hope of buying them back later at a lower price.
- ☐ **Covering (closing) a short position:** buying shares to return to the lender, which ends the short position.
- ☐ **Initial margin:** the minimum equity, as a percentage of position value, required to open the position.
- ☐ **Maintenance margin:** the minimum equity percentage that must be kept; falling below it triggers a margin call.
- ☐ **Margin call:** the broker's demand for more cash or securities, or a closed position, when margin falls below maintenance.
- ☐ **Short interest:** the number of shares of a stock currently sold short.
- ☐ **Short squeeze:** a rapid price rise that forces short sellers to cover, which pushes the price higher still.
- ☐ **Stop-buy order:** an order to buy if the price rises to a stated level; used to cap losses on a short sale.
- ☐ **Uptick rule / alternative uptick rule (SEC Rule 201):** once a stock falls 10% or more in a day, short sales are allowed only above the current national best bid for the rest of that day and the next.

CFA-style practice questions

Q1 [Algorithmic]. An investor shorts 2,000 shares at \$60 with 50% initial margin and 30% maintenance margin. The price at which she receives a margin call is *closest* to:

- A. \$42.86
- B. \$69.23
- C. \$78.00

Q2 [Algorithmic]. Using the position in Q1, the stock rises to \$72. Her margin percentage is *closest* to:

- A. 20.0%
- B. 25.0%
- C. 33.3%

Q3 [Algorithmic]. At \$72, how much cash must she deposit to restore the account to the 30% maintenance margin?

- A. \$7,200
- B. \$36,000
- C. \$43,200

Q4 [Algorithmic]. An investor shorts 1,000 shares at \$50 with 50% initial margin. The stock pays a \$0.50 dividend and she covers at \$42. Her return on the margin deposit is *closest* to:

- A. 15%
- B. 30%
- C. 32%

Q5 [Conceptual]. The potential loss on a short sale is theoretically unlimited because:

- A. the stock price has no upper limit.
- B. the broker can charge unlimited interest on borrowed shares.
- C. the short seller must pay all future dividends.

Q6 [Algorithmic]. A short sale is opened at \$100 with 50% initial margin. If maintenance margin is 40% instead of 30%, the margin call price is *closest* to:

- A. \$107.14
- B. \$115.38
- C. \$125.00

Q7 [Conceptual]. The proceeds from a short sale:

- A. may be withdrawn to buy other securities.
- B. must stay in the investor's margin account.
- C. are paid to the investor whose shares were borrowed.

Q8 [Conceptual]. When a stock that has been sold short pays a cash dividend:

- A. the short seller receives the dividend.
- B. the short seller must pay the dividend to the lender of the shares.
- C. no one receives the dividend on the borrowed shares.

Answer key

Q	Answer	Explanation
1	B	Proceeds \$120,000 + deposit \$60,000 = \$180,000. $P^* = 180,000 / (2,000 \times 1.30) = \69.23 . A applies the long formula; C is 60×1.30 .
2	B	Shares owed = \$144,000. Equity = $180,000 - 144,000 = \$36,000$. Margin = $36,000 / 144,000 = 25\%$.

Q	Answer	Explanation
3	A	Required equity = $0.30 \times 144,000 = \$43,200$. Deposit = $43,200 - 36,000 = \$7,200$. C is the total required equity, not the shortfall. B restores to 50%.
4	B	Gain = $(50 - 42) \times 1,000 - \500 dividend = $\$7,500$. Deposit = $\$25,000$. Return = 30%. C ignores the dividend (32%); A divides by the $\$50,000$ proceeds.
5	A	A stock price can rise without limit, so the cost to cover has no ceiling.
6	A	$P^* = 100 \times 1.50 / 1.40 = \107.14 . A higher maintenance margin triggers the call sooner (at a lower price). B uses 30%.
7	B	Proceeds stay with the broker as collateral for the borrowed shares.
8	B	The lender of the shares is entitled to the dividend, so the short seller must pay it.